Review plan · for the teacher

Computational thinking, unplugged and with blocks: unit check

Programming · Level 1 · Unit check

Time
45 min

Materials
Board, chalk, paper. No devices needed.

Learning goals

- Algorithms: exact instructions
- Loops
- Decisions and debugging
- Break it down

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Act out the robot questions on a chalk grid

Discussion

- Which game will you play with your class first?
- Which question surprised you, and why?
- Which idea will your pupils find hardest?

Exit ticket

- One unplugged activity I will run, and when

Try it in class tomorrow
Pick one game from this course (the human robot, clap the loop, Simon says IF or guess my number) and play it with a class this week. Then tell a colleague how it went.

Quiz sheet · for each team

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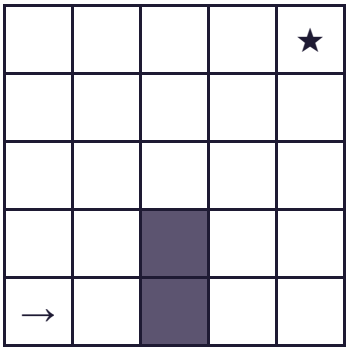
Team _____ Date _____

1 **What is an algorithm?**

◯ A guess that is usually right ◯ A computer program that is very fast ◯ A kind of computer

◯ Exact steps, in order, that finish and solve a problem

2 **Walk the robot around the wall to the flag, with 5 cards at most.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

3 **A loop makes a program longer, but easier to read.**

◯ True ◯ False

4 **Match each part of computational thinking to an example.**

Decomposition	• • • •	Noticing the canteen queue is longest every Monday
Pattern recognition	• • • •	Planning sports day as events, teams and timings
Abstraction	• • • •	A recipe for bread
Algorithm	• • • •	A timetable that shows subjects, not what is taught

5 **The robot starts facing up and runs this program. Which way is it facing at the end?**

```
for i in range(3):
    forward()
    right()
```

◯ Up ◯ Down ◯ Left ◯ Right

6 **The school rule is: "go out at break, unless it is raining". Which line of this program is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  if break_time and raining:
2      go_out()
3  else:
4      stay_in()
```

---

7 **Put this algorithm for finding the tallest pupil in a row in order.**

- [ ] The next pupil stands back to back with them
- [ ] The first pupil stands at the front
- [ ] Repeat until everyone has been compared
- [ ] The taller one stays at the front

8 **Which are good debugging habits?**

- [ ] Rewrite everything when it fails
- [ ] Find the first step that goes wrong
- [ ] Change one thing at a time
- [ ] Test again after every change

9 **Complete the sentence.**

Breaking a problem into parts is _____, and keeping only what matters is _____.

Word bank: abstraction · debugging · decomposition · a loop

10 **With halving, what is the most guesses you need for a number from 1 to 100?**

- ◯ 7
- ◯ 50
- ◯ 100

11 **Rule: "IF it is cold OR it is raining, wear a coat." On a cold, dry day, you wear a coat.**

- ◯ True
- ◯ False

Answers · for the teacher

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1 **Exact steps, in order, that finish and solve a problem**

An algorithm is a precise list of steps, in order, that finishes.

2 **Any program that takes the robot to the ★ with 5 cards or fewer.**

Turn left, forward × 4, turn right, forward × 4: four cards, using repeats for the long straight lines.

3 **False**

A loop makes a program shorter and easier to read, and easier to change: one number instead of many repeated steps.

4 **Decomposition — Planning sports day as events, teams and timings**
Pattern recognition — Noticing the canteen queue is longest every Monday
Abstraction — A timetable that shows subjects, not what is taught
Algorithm — A recipe for bread

Parts of a big job, something that repeats, only what matters, and exact steps: the four ideas behind every program.

5 **Left**

Three right turns: up, then right, then down, then left.

6 **Line 1: `if break_time and not raining:`**

The condition sends pupils out only when it is raining: the opposite of the rule.

7 **Top to bottom: 2, 1, 4, 3**

Keep the tallest so far and compare each new pupil with them: one pass through the row.

8 **Find the first step that goes wrong; Change one thing at a time; Test again after every change**

The routine is: find the first wrong step, change one thing, test again.

9 **Breaking a problem into parts is <u>decomposition</u>, and keeping only what matters is <u>abstraction</u>.**

Decomposition splits a problem into parts; abstraction leaves out the details that don't matter for the job.

10 **7**

Each guess halves what is left: 100, 50, 25, 13, 7, 4, 2, 1.

11 **True**

With OR, one true part is enough. It is cold, so the whole condition is true, even though it isn't raining.